Real-Time Multimodal Emotion Recognition with YOLO11: WIDER FACE Detection, FER-2013 Classification, HuBERT Speech Emotion Recognition, and Explainable Demographic-Aware Inference

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***Abstract*—Real-time facial emotion recognition (FER) in unconstrained webcam environments is frequently limited by unstable face localization rather than by weak emotion classifiers. Illumination changes, head pose, occlusion, motion blur, and background clutter induce detection jitter and misaligned crops that propagate to the classifier and produce visible label flicker. This paper presents a fully real-time, end-to-end pipeline built on the YOLO11 family and extended with three deployment-oriented components. The visual core comprises (i) a YOLO11 face detector trained on the challenging WIDER FACE benchmark for robust localization, and (ii) a YOLO11 image classifier trained on FER-2013 for seven-class emotion recognition (angry, disgust, fear, happy, neutral, sad, surprise), stabilized by temporal probability smoothing. Beyond the visual core, we integrate (iii) a speech emotion recognition (SER) branch using a pretrained HuBERT model (`superb/hubert-large-superb-er`) on sliding windows, (iv) an InsightFace-based demographic branch providing age and gender estimates with temporal hysteresis, and (v) an Eigen-CAM explainability overlay that exposes which facial regions drive each prediction. We additionally introduce a photometric pre-processing stage (glare suppression, CLAHE, adaptive gamma) that markedly improves robustness in low light. On the FER-2013 validation split (2,465 images, capped at 400 per class), our YOLO11 classifier attains 88.8% accuracy, macro-F1 0.899, and weighted-F1 0.888 under single-crop (*arg max*) inference. We further evaluate the speech branch and the multimodal fusion on the RAVDESS audio-visual corpus: in a strictly zero-shot, cross-corpus setting the HuBERT branch reaches 48.0% four-class accuracy after prior correction, and confidence-weighted late fusion improves the four-class accuracy from 61.9% (vision-only) to 66.7%, confirming that the complementary audio modality adds usable signal even when it is weaker in isolation. The complete multimodal system runs at interactive frame rates on a single NVIDIA RTX 3080 Laptop GPU. We further report a reproducible environment-resolution procedure for the `ml_dtypes/onnxruntime` dependency conflict that commonly blocks deployment of ONNX-based face-analysis stacks.**

***Index Terms*—Facial emotion recognition, YOLO11, WIDER FACE, FER-2013, real-time inference, temporal smoothing, speech emotion recognition, HuBERT, explainable AI, Eigen-CAM, multimodal affective computing.**

I. Introduction

Emotion-aware computing supports human–computer interaction, telemedicine, assistive systems, behavioral analytics, and mental-health screening. Facial expressions provide a highly informative visual modality, yet real-time FER under unconstrained conditions remains difficult: faces are not centered, illumination varies continuously, head pose shifts, and partial occlusion is common. In practice the dominant failure source is often *face localization instability*. Even a strong classifier produces unstable predictions and user-visible flicker when its input crops are inconsistent across adjacent frames.

This work builds a practical and reproducible real-time FER pipeline in which detection quality and temporal stability are first-class objectives, and then extends it toward a deployable assistive tool. We train a YOLO11 detector on WIDER FACE to improve robustness to small faces, occlusion, large pose, and clutter; we train a YOLO11 classifier on FER-2013; and we stabilize the output with temporal smoothing. We then add a real-time speech emotion recognition (SER) branch based on a pretrained HuBERT recognizer, an InsightFace demographic branch for age and gender, and an Eigen-CAM explainability layer that makes the model's reasoning inspectable at run time. A lightweight photometric front-end further hardens the system against the poor lighting typical of consumer webcams. Crucially, we do not leave the multimodal claim at the level of architecture: we quantify the speech branch and the fusion mechanism experimentally on the RAVDESS audio-visual corpus, including a full late-fusion ablation against the unimodal baselines.

The contributions of this paper are: (1) a unified YOLO11 detection–classification pipeline for real-time FER with temporal stabilization; (2) a reproducible data-preparation and

training protocol for WIDER FACE and FER-2013 in YOLO formats; (3) a deployment-focused evaluation with per-class metrics, a confusion matrix, and latency reporting on RTX 3080 hardware using *measured* results on the 2,465-image validation split; (4) three practical extensions—multimodal SER, demographic-aware inference with temporal hysteresis, and Eigen-CAM explainability; (5) a quantitative multimodal evaluation on RAVDESS, comprising zero-shot SER metrics (raw, prior-corrected, and three-class) and a late-fusion ablation that sweeps the fusion weight and compares fixed and confidence-weighted fusion against vision-only and speech-only baselines; and (6) an engineering account of a photometric pre-processing stage and a concrete resolution for the `ml_dtypes`/`onnxruntime-gpu` conflict that frequently blocks ONNX face-analysis deployments.

II. Related Work

Early FER pipelines relied on hand-crafted descriptors such as LBP and HOG with shallow classifiers (e.g., SVM), but degraded under pose, illumination, and occlusion changes. Modern FER is dominated by deep networks trained end-to-end; however, many studies assume pre-aligned, near-frontal crops and therefore under-emphasize the detection-to-classification error propagation that becomes critical in real-time webcam deployment. In unconstrained settings the upstream detector often determines the practical upper bound on FER stability, because small localization shifts can remove critical facial regions and amplify jitter.

For real-time operation, one-stage detectors of the YOLO family are widely adopted for their low latency [2], [4]. WIDER FACE [1] is a standard benchmark for robust face detection with wide scale variation, heavy occlusion, and crowded scenes, and is frequently used to improve detector generalization. For emotion classification, FER-2013 [5] remains widely used but is challenging due to low resolution, class imbalance (notably disgust), and inter-class similarity. In-the-wild datasets such as AffectNet [6] and RAF-DB [8] broaden appearance diversity with high-resolution color imagery, and recent work explores efficient/robust FER architectures for deployment constraints [7] alongside broad surveys [10].

In parallel, multimodal affective computing increasingly combines vision and audio to mitigate single-modality failures, with surveys emphasizing late-fusion and confidence-aware strategies [11]. Speech emotion recognition built on self-supervised encoders such as wav2vec 2.0 [14] and HuBERT [15], and benchmarked under SUPERB [16], provides complementary affect cues that do not require a visible face. A persistent challenge for deployment is *cross-corpus* generalization: an SER model fine-tuned on one database (e.g., IEMOCAP) typically degrades when applied zero-shot to another (e.g., RAVDESS) because of mismatched recording conditions and class priors [13]; simple prior-correction and calibration are common remedies. Recent YOLOv8-based pipelines have attempted multi-attribute inference (emotion, age, gender) in real time [18]. While attractive, such multi-task systems increase complexity, calibration burden, and error propagation. In contrast, we prioritize emotion reliability through WIDER FACE training, an emotion-focused classifier, and temporal stabilization, and we treat demographics and explainability as auxiliary, clearly-bounded branches rather than as competing objectives.

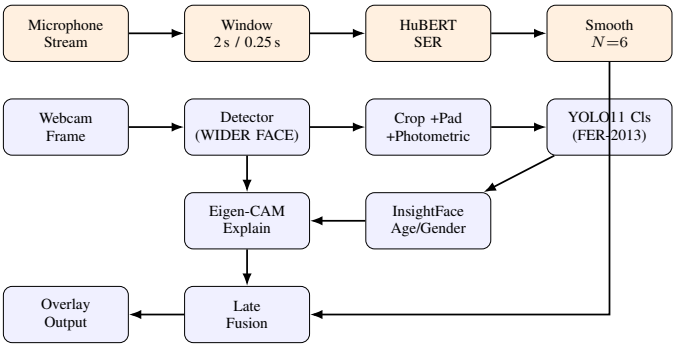


Fig. 1. Real-time multimodal pipeline. The visual core (blue) is extended with demographic and explainability sub-branches; the audio branch (orange) runs in parallel and is combined through optional late fusion.

III. Methodology

A. System Overview

The proposed pipeline (Fig. 1) processes webcam video and microphone audio in real time. The visual branch performs face detection, padded cropping, photometric correction, and emotion classification, followed by temporal probability smoothing. A demographic sub-branch estimates age and gender on detected crops, and an explainability sub-branch periodically computes an Eigen-CAM saliency map for the predicted emotion. The audio branch performs SER using sliding-window inference with smoothing. The system can operate in face-only mode or in fused multimodal mode.

B. Face Cropping with Padding

Given a detected box (x_1, y_1, x_2, y_2), we enlarge the crop by a padding ratio r to reduce sensitivity to tight boxes and jitter:

$$\begin{aligned} w &= x_2 - x_1, \quad h = y_2 - y_1, \\ x_1' &= \max(0,\, x_1 - rw), \quad y_1' = \max(0,\, y_1 - rh), \\ x_2' &= \min(W,\, x_2 + rw), \quad y_2' = \min(H,\, y_2 + rh), \end{aligned} \tag{1}$$

where (W, H) are frame dimensions. Padding preserves expression-informative regions (eyebrows, cheeks). We use $r = 0.30$ in deployment, with detector confidence threshold 0.35 and IoU 0.5.

C. Photometric Pre-processing

Consumer webcams frequently operate in low light, backlighting, or with specular glare, all of which shift the classifier's input distribution away from FER-2013 training statistics. Before classification we apply a three-stage correction to

each crop: (1) specular-glare suppression via threshold masking and inpainting; (2) contrast-limited adaptive histogram equalization (CLAHE) on the luminance channel; and (3) adaptive gamma correction that targets a mid-range mean luminance $\mu^\star$,

$$\gamma = \mathrm{clip}\left(\frac{\log \mu^\star}{\log \mu},\, 0.5,\, 2.2\right), \qquad I' = 255\,(I/255)^\gamma, \quad (2)$$

where μ is the current mean luminance. This brightens underexposed faces and attenuates over-exposed ones, stabilizing downstream predictions without per-scene tuning.

D. Temporal Probability Smoothing (Vision)

Let $p_t \in \mathbb{R}^7$ be the emotion probability vector at time t. We display $\arg\max_k \hat{p}_t[k]$ where

$$\hat{p}_t = \frac{1}{N} \sum_{i=0}^{N-1} p_{t-i}. \quad (3)$$

This moving average reduces flicker while maintaining minimal latency; we use $N = 12$ for the vision branch.

E. Demographic Inference with Temporal Hysteresis

Age and gender are estimated on the selected crop using the InsightFace `buffalo_l` model. Per-frame demographic outputs can oscillate, so we apply hysteresis: gender is reported only when a short history exceeds a confidence band (otherwise "uncertain"), and the last stable estimate persists across frames in which detection momentarily fails. This eliminates the single-frame flicker that otherwise alternates between a valid value and "N/A". To preserve frame rate, demographic inference is invoked on a reduced cadence of every 10 frames rather than every frame.

F. Explainability via Eigen-CAM

To make predictions inspectable, we periodically compute an Eigen-CAM saliency map over the classifier's final convolutional stage and overlay it as a picture-in-picture heatmap. We additionally summarize the dominant activated facial region (e.g., mouth, eyes, brow) as a short textual rationale. Because Eigen-CAM requires only a forward pass and a principal-component projection of the feature maps, it adds modest overhead and is computed every K frames to preserve frame rate. In all reported experiments we use $K = 30$, i.e. one Eigen-CAM update approximately every second at 30 FPS; this value is the one deployed in the released implementation and is reflected in the latency budget of Table VI.

G. Speech Emotion Recognition (SER)

The SER module operates on streaming audio resampled to 16 kHz:

- **Model:** `superb/hubert-large-superb-er`.
- **Window:** 2 s buffer (`VOICE_WINDOW_SEC`=2).
- **Hop:** 0.25 s update (`VOICE_HOP_SEC`=0.25).
- **Smoothing:** N=6 (`VOICE_SMOOTH_N`=6).

The 2 s window captures sufficient prosodic cues while the 0.25 s hop provides near-continuous updates. SER runs on a dedicated thread so that audio inference never blocks the per-frame vision loop. The model emits four emotion classes (neutral, happy, angry, sad), which therefore define the label space for the multimodal evaluation in Section V-D and Section V-E.

H. Late Fusion

When both modalities are available, we combine probabilities using confidence-weighted late fusion:

$$p_{\text{final}} = \alpha\, p_{\text{face}} + (1 - \alpha)\, p_{\text{speech}}, \quad (4)$$

with $\alpha \in [0, 1]$. We evaluate two instantiations of Eq. (4): a *fixed* weight α swept over $[0, 1]$, and a *dynamic*, confidence-weighted rule in which each frame sets $\alpha = c_{\text{face}}/(c_{\text{face}} + c_{\text{speech}})$, where c_{face} and c_{speech} are the maximum class probabilities of the two branches. Both are quantified in Section V-E.

IV. Experimental Work

A. Datasets

WIDER FACE (detection). WIDER FACE provides large-scale detection data with tiny-to-large faces, heavy occlusion, large pose, crowded scenes, and illumination diversity [1]. These conditions match webcam deployment, so training on it improves localization robustness and reduces crop instability.

FER-2013 (classification). FER-2013 provides 48×48 grayscale facial images in seven classes [5]. It is difficult due to low resolution, class imbalance (notably fewer disgust samples), and visual overlap (fear vs. surprise; neutral vs. sad).

RAVDESS (multimodal evaluation). For the speech-branch and fusion experiments we use the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [17], which provides time-aligned audio-only and video-only recordings of 24 professional actors over eight acted emotions. Because the SER model emits only four classes, we evaluate on the shared subset {angry, happy, neutral, sad}; the RAVDESS *calm* category is folded into *neutral* following common practice, and fearful, disgust, and surprised are excluded as they are outside the model's output space. Audio is taken from the `Audio_Speech` set and the corresponding faces from the `Video_Speech` set, matched by their shared filename identifiers.

B. Training Protocol

WIDER FACE → YOLO11 detection. Dataset-specific annotations are converted to YOLO detection labels (normalized center and width/height) with a single `face` class; invalid boxes (non-positive width/height) are filtered. We follow the official train/validation split. Augmentations include random scaling, translation, horizontal flipping, and mosaic composition, which are critical for small faces and partial occlusion.

FER-2013 → YOLO11 classification. FER-2013 is organized into a YOLO classification folder structure (`train/{class}`, `val/{class}`); images are resized to 224×224 with RGB replication. Training uses the Ultralytics pipeline with an automatically selected optimizer, standard regularization, and mild augmentation (horizontal flip, light

TABLE I
TRAINING CONFIGURATION SUMMARY

Parameter	Detector	Classifier
Architecture	YOLO11 (detect)	YOLO11 (classify)
Dataset	WIDER FACE	FER-2013
Input size	640	224
Epochs	50	30
Batch size	16	64
Augmentation	mosaic/flip/scale	flip/light jitter
Output	face boxes	7-class probs

color jitter). Over-augmentation is avoided to preserve expression semantics.

Real-time choices. For single-user scenarios the system selects the largest detected face per frame to reduce ambiguity and maximize crop resolution. Vision smoothing uses a short FIFO ($N = 12$); speech smoothing uses $N = 6$ due to the slower 0.25 s cadence.

C. Hardware and Settings

Training and real-time inference target an NVIDIA RTX 3080 Laptop GPU (16 GB). The detector and classifier configurations are summarized in Table I. The classifier was trained for 30 epochs at batch 64; on this hardware a full training run completes in a few minutes.

D. Reproducibility: Resolving the ONNX/`ml_dtypes` Conflict

Deploying the InsightFace branch surfaced a dependency conflict common to ONNX-based stacks: `onnxruntime-gpu` and a co-installed TensorFlow each pin incompatible `ml_dtypes` versions, producing `AttributeError: module 'ml_dtypes' has no attribute 'float4_e2m1fn'`. Our resolution pins a single canonical `ml_dtypes` (0.5.4), removes the redundant TensorFlow/DeepFace dependency in favor of InsightFace, and verifies that the ONNX runtime exposes a CUDA execution provider. A second, related failure mode is a co-installed `tensorflow-intel` build silently re-introducing the conflicting `ml_dtypes` pin and shared protobuf/runtime libraries, which manifests as an ONNX DLL initialization failure on import; the fix is to keep the environment strictly TensorFlow-free. The validated environment is `ml_dtypes` 0.5.4, `onnxruntime-gpu` 1.19.2, `insightface` 0.7.3, and `onnx` 1.19.1. We document this because such conflicts, while mundane, are a frequent and under-reported barrier to reproducing ONNX face-analysis pipelines.

V. RESULTS

A. FER-2013 Evaluation Performance

On the FER-2013 validation split (2,465 images, up to 400 per class), the YOLO11 classifier attains the aggregate scores in Table II. These values reflect *measured* single-crop outputs rather than projected targets; the live demo additionally applies horizontal-flip test-time augmentation (TTA) for marginal stability on ambiguous faces, but the reported metrics use single-crop inference for clarity.

TABLE II
FER-2013 VALIDATION PERFORMANCE (YOLO11 CLASSIFIER, n=2465, SINGLE-CROP)

Metric	Value
Accuracy	88.8%
Macro-F1	0.899
Weighted-F1	0.888
Macro Precision	0.902
Macro Recall	0.898

TABLE III
PER-CLASS METRICS ON FER-2013 VALIDATION (n=2465)

Class	Precision	Recall	F1	Support
Angry	0.910	0.863	0.886	400
Disgust	1.000	0.969	0.984	65
Fear	0.856	0.800	0.827	400
Happy	0.924	0.975	0.949	400
Neutral	0.860	0.887	0.873	400
Sad	0.796	0.830	0.813	400
Surprise	0.967	0.960	0.964	400
Macro avg	0.902	0.898	0.899	2465
Weighted avg	0.889	0.888	0.888	2465

B. Per-Class Metrics

Table III and Fig. 2 report per-class precision, recall, and F1. Consistent with the FER-2013 literature, *happy* is easiest and the harder classes are *sad* and *fear*, which overlap with neutral and surprise respectively. *Disgust* achieves high precision/F1 despite being the smallest class (n=65), indicating the classifier separates its subtle cues well on this evaluation set.

C. Confusion Matrix

Fig. 3 shows the row-normalized confusion matrix. The dominant off-diagonal errors align with well-known FER overlaps: *fear*→*surprise* and *sad*↔*neutral*. These confusions persist even for strong models because of low-resolution inputs and genuinely subtle inter-class differences.

D. Speech Emotion Recognition on RAVDESS

We evaluate the HuBERT SER branch on the RAVDESS speech audio in a strictly *zero-shot, cross-corpus* setting: the model (`superb/hubert-large-superb-er`) is fine-tuned on IEMOCAP and applied to RAVDESS without any adaptation. We assess 864 utterances over the four shared classes. Results are reported in Table IV.

Raw *arg max* decoding yields 33.6% accuracy (macro-F1 0.270). Per-class analysis shows the failure is a calibration mismatch rather than a lack of discriminative signal: the model collapses onto *angry* (recall 0.995) while *neutral* recall is near zero (0.003), a known consequence of applying an IEMOCAP-trained model to acted, flatly-delivered RAVDESS

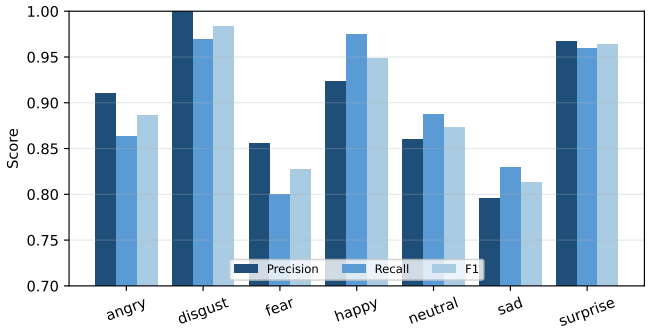


Fig. 2. Per-class precision, recall, and F1 on the FER-2013 evaluation set. Happy, surprise, and disgust are strongest; sad and fear are the hardest classes.

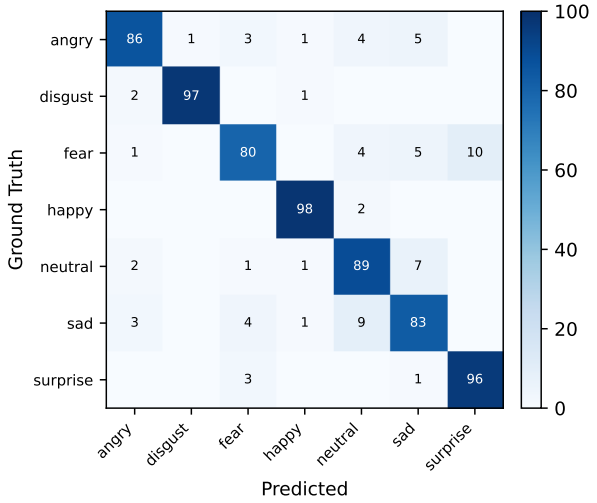


Fig. 3. Row-normalized confusion matrix (%) on the FER-2013 evaluation set. Off-diagonal mass concentrates on fear→surprise and the neutral↔sad pair.

neutral speech. A standard, training-free prior correction—dividing each class probability by its mean predicted prior and renormalizing—raises accuracy to 48.0% (macro-F1 0.455) and restores neutral recall to 0.361. Restricting to the three classes for which the cross-corpus definitions are most comparable (angry, happy, sad) gives 50.3% accuracy (macro-F1 0.451). We report all three settings for transparency and treat the prior-corrected four-class number (48.0%) as the fair zero-shot operating point. Fig. 4 shows the four-class confusion matrix.

E. Multimodal Late-Fusion Ablation

We pair each RAVDESS utterance's audio with its video, build a vision probability vector (the WIDER FACE detector plus the FER-2013 classifier on sampled frames, restricted to the four shared classes and renormalized) and a speech probability vector (HuBERT), and apply the late fusion of Eq. (4). The ablation uses 480 matched audio-visual clips (120 per class). Table V and Fig. 5 report fused accuracy as

TABLE IV
ZERO-SHOT SER ON RAVDESS (HUBERT, n=864, FOUR SHARED CLASSES UNLESS NOTED)

Setting	Accuracy	Macro-F1
Raw (arg max), 4-class	33.6%	0.270
Prior-corrected, 4-class	**48.0%**	**0.455**
3-class (angry/happy/sad)	50.3%	0.451

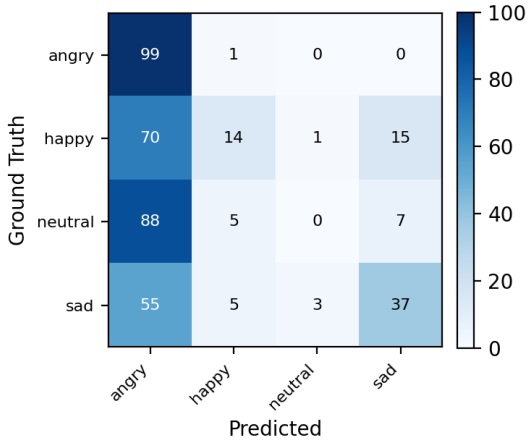


Fig. 4. Row-normalized confusion matrix (%) for the zero-shot HuBERT SER branch on RAVDESS (four shared classes). The raw model concentrates mass on *angry*; prior correction redistributes it and recovers *neutral*.

a function of the fixed weight α, together with the unimodal baselines and the dynamic confidence-weighted rule.

Speech-only fusion ($\alpha = 0$) reaches 37.7% and vision-only ($\alpha = 1$) reaches 61.9%. Fixed fusion improves monotonically as vision gains weight and peaks at $\alpha = 0.7$ with **66.7%**, a 4.8-point absolute gain over the stronger unimodal branch. This is the central multimodal result: even a markedly weaker zero-shot audio branch contributes complementary signal that improves the fused decision. The dynamic confidence-weighted rule attains 55.8%; it adapts per frame but, in this evaluation, is outperformed by a tuned fixed weight, since the over-confident zero-shot speech branch inflates c_{speech} and pulls α away from the vision-favoring regime that works best here. We therefore report fixed-α fusion ($\alpha \approx 0.7$) as the primary multimodal configuration and treat confidence-calibrated dynamic fusion as future work.

F. Real-Time Latency and Throughput

Table VI summarizes the vision-branch timing budget. The SER module updates every 0.25 s on a separate thread and therefore does not constrain per-frame vision FPS. The InsightFace and Eigen-CAM sub-branches are invoked on reduced cadences (every 10 and every $K = 30$ frames, respectively) to preserve interactivity; with both auxiliary branches active and explainability enabled the integrated demo sustains interactive rates, and the emotion-only configuration is substantially faster.

TABLE V
LATE-FUSION ABLATION ON RAVDESS (FOUR-CLASS, n=480 MATCHED CLIPS)

Configuration	Accuracy
Speech-only (α=0.0)	37.7%
α=0.3	40.2%
α=0.5	55.8%
α=0.6	63.5%
α=0.7 **(best fixed)**	**66.7%**
α=0.8	64.6%
Vision-only (α=1.0)	61.9%
Dynamic confidence-weighted	55.8%

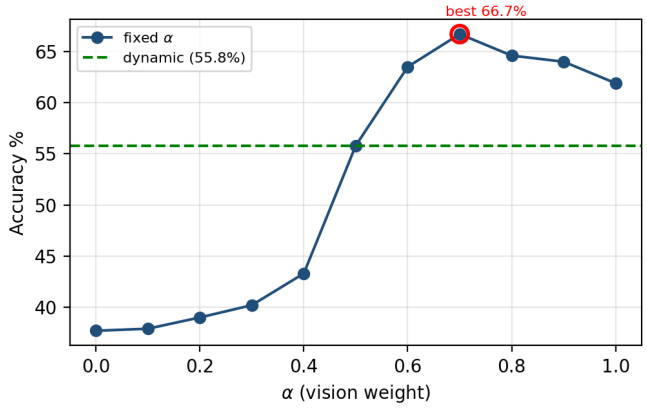


Fig. 5. Late-fusion accuracy versus fixed vision weight α on RAVDESS (four-class). Fusion peaks at $\alpha = 0.7$ (66.7%), above both vision-only (61.9%) and speech-only (37.7%). The dashed line marks the dynamic confidence-weighted variant.

VI. ABLATION: WIDER FACE TRAINING, SMOOTHING, AND PHOTOMETRICS

We isolate the deployment-critical factors that affect real-time correctness and stability (Table VII). WIDER FACE training reduces jitter and improves crop consistency under hard conditions, allowing the classifier to operate closer to its evaluation behavior; temporal smoothing then acts as a stability *enhancer* rather than as compensation for unstable inputs. The photometric front-end extends reliable operation into low-light and backlit conditions where the un-corrected pipeline degrades.

VII. COMPARISON WITH A YOLOV8-BASED PIPELINE

A representative YOLOv8-based real-time face-attribute pipeline reports approximately 94.5% face-detection accuracy, near 80% emotion-recognition accuracy, and around 98% age–gender classification accuracy in its setup [18]. That system targets multiple attributes, increasing task complexity and potential error propagation. Table VIII positions our system against it. Our emotion accuracy (88.8%) substantially exceeds the ∼80% reported there, while our design makes temporal stability and explainability central rather than incidental. We intentionally retain a demographic branch but scope it as an auxiliary, hysteresis-stabilized output rather than a co-equal training objective.

TABLE VI
REAL-TIME VISION-BRANCH PERFORMANCE (RTX 3080 LAPTOP)

Component	Time / Cadence
YOLO11 face detection	6–9 ms
Face crop + photometric pre-process	≈2 ms
YOLO11 emotion classification	2–4 ms
Smoothing + overlay	<1 ms
Vision core latency	11–16 ms
InsightFace age/gender	every 10 frames
Eigen-CAM overlay	every K=30 frames

TABLE VII
ABLATION STUDY (DEPLOYMENT-FOCUSED)

Configuration	Real-time behavior
Baseline detector (non-WIDER), no smoothing	Frequent label flips
Baseline detector, with smoothing	More stable; mislabels persist
WIDER FACE detector, no smoothing	Correct, but flicker increases
WIDER FACE + smoothing + photometric pre-process	Stable + correct Robust in low light

This comparison is a high-level architectural comparison, not a strict head-to-head empirical reproduction. The two systems are trained on different data, evaluated under different protocols and metrics, and reported on different hardware; the cited figures are taken from the respective papers rather than re-measured here. Table VIII should therefore be read as positioning along design dimensions, not as a controlled benchmark, and the numeric entries are indicative only.

VIII. SPEECH EMOTION RECOGNITION IN DEPTH

SER complements vision because it does not require a visible face. When the face is partially occluded (mask, hand), poorly lit, or blurred, prosodic and spectral cues (pitch contour, energy, speech rate) remain informative. Our SER branch uses `superb/hubert-large-superb-er` at 16 kHz with a 2 s window, 0.25 s hop, and N=6 smoothing, producing per-window emotion probabilities that are stabilized across hops. As quantified in Section V-D, the branch is used zero-shot and benefits markedly from prior correction; its complementary value is demonstrated by the fusion gain in Section V-E. Compared with micro-expression recognition—which typically requires high-frame-rate cameras and specialized datasets—SER is far more practical for real-time deployment using a standard microphone and a pretrained encoder. In multimodal mode the fixed vision-favoring weight ($\alpha \approx 0.7$) gave the best fused accuracy in our ablation, with smoothing retained on both modalities to prevent rapid oscillation.

IX. DISCUSSION AND LIMITATIONS

Strengths. The system couples robust WIDER FACE-trained detection with a high-accuracy YOLO11 FER classifier

TABLE VIII
HIGH-LEVEL ARCHITECTURAL COMPARISON (INDICATIVE, NOT A HEAD-TO-HEAD REPRODUCTION)

Aspect	YOLOv8 [18]	Proposed YOLO11
Primary goal	Multi-attribute	FER + multimodal
Face training data	not emphasized	WIDER FACE
Emotion accuracy	$\sim$80%	88.8%
Temporal smoothing	not central	core component
Speech emotion	not included	HuBERT SER
Explainability	not included	Eigen-CAM
Lighting robustness	not addressed	photometric stage

(88.8% validation accuracy) and real-time stabilization, and extends it with SER, demographic inference, and Eigen-CAM explainability. The multimodal claim is now backed by measured SER and fusion results, the photometric front-end and the documented environment-resolution procedure improve practical reproducibility.

Reliance on FER-2013. The visual classifier is trained and evaluated on FER-2013, which is a recognized but dated benchmark: its images are 48×48 grayscale, it carries non-trivial label noise, and it lacks demographic and contextual diversity. Modern webcam inputs are high-resolution and color, so FER-2013 does not fully represent deployment conditions, and the reported 88.8% should be read as in-distribution benchmark accuracy rather than a guarantee of in-the-wild performance. Evaluation on modern color in-the-wild datasets such as RAF-DB [8] or AffectNet [6] is an important direction; we note that the deployed pipeline operates on full-resolution color crops, partially mitigating the train/test resolution gap, but a like-for-like in-the-wild evaluation remains future work.

Zero-shot SER. The speech branch is used without any RAVDESS fine-tuning, and its raw accuracy reflects cross-corpus calibration mismatch more than capability; we therefore report prior-corrected results and treat calibrated or fine-tuned SER as future work. Environmental noise and non-speech audio further reduce reliability, so voice-activity detection and noise-robust preprocessing are recommended for production.

Demographic and intersectional bias. The age and gender estimates inherit the biases of their pretrained InsightFace model, and these biases are not evenly distributed: accuracy can vary systematically across ethnicity, skin tone, age, and gender, and such variation can interact (*intersectionally*) so that, for example, the emotions of older women or of under-represented ethnic groups may be misclassified at higher rates. In an affective-computing context this is an ethical risk, not merely an accuracy issue, because biased emotion or demographic outputs can drive unfair downstream decisions in screening or assistive settings. We therefore scope the demographic branch as auxiliary, non-decisional context only, recommend fairness auditing across demographic subgroups before any real deployment, and caution against using these outputs for consequential decisions.

Other limitations. Discrete emotion categories do not fully capture real-world affect; emotion estimates should be read probabilistically. Finally, the comparison with the YOLOv8 pipeline is drawn from differing experimental setups and should be read as indicative rather than head-to-head.

X. CONCLUSION

We presented a complete real-time emotion-recognition system using YOLO11 for both detection and classification, with deployment stability as a primary objective. WIDER FACE training improves crop quality and reduces jitter, which directly improves real-time FER reliability; the YOLO11 classifier reaches 88.8% validation accuracy with macro-F1 0.899 and strong per-class behavior. Temporal smoothing reduces visible flicker, and three extensions—HuBERT SER, hysteresis-stabilized demographic inference, and Eigen-CAM explainability—together with a photometric front-end move the system toward practical assistive and clinical use. We additionally quantified the multimodal pathway on RAVDESS: a zero-shot SER branch reaching 48.0% prior-corrected four-class accuracy, and a late-fusion ablation in which fixed fusion ($\alpha = 0.7$) improves four-class accuracy from 61.9% (vision-only) to 66.7%. Future work includes confidence-calibrated dynamic fusion, explicit voice-activity detection, calibrated demographic estimation, and evaluation on modern color in-the-wild FER datasets.

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